🚢 SmartyShips: Technical Evaluation Report

Freight Transit Duration Forecasting, SLA Risk Assessment & Delay Propagation

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1. Executive Summary & Problem Formulation

Freight logistics networks face compounding uncertainties across long-haul highway lanes, break-bulk cross-dock sorting hubs, and last-mile courier routes. Inaccurate transit time estimates lead to warehouse dock congestion, missed customer Service Level Agreements (SLAs), and costly downstream expediting.

SmartyShips is an end-to-end machine learning system and operational dashboard built to address three specific supply chain capabilities:

- **Pre-Shipment Duration Forecasting:** Supervised regression estimating delivery duration from attributes strictly available prior to departure.
- **SLA Risk Classification:** Auditing predicted arrival times against contract deadlines into standardized operational risk tiers (*On Time*, *Low Risk*, *Moderate Risk*, *High Risk*).
- **Multi-Stage Disruption Propagation:** A single-ledger simulation engine modeling how upstream delays cascade across downstream transit legs without recursive double-counting.

2. Dataset Audit & Data Hygiene

2.1 Source Data & Attribution

The platform utilizes the **US Logistics Performance Dataset on Kaggle**, published by **Shahriar Kabir**. The raw dataset contains 2,000 shipment records across 11 features: `Shipment_ID`, `Origin_Warehouse`, `Destination`, `Carrier`, `Shipment_Date`, `Delivery_Date`, `Weight_kg`, `Cost`, `Status`, `Distance_miles`, and `Transit_Days`.

> **Data Scope & Provenance Notice:** The raw dataset records pre-departure attributes and calendar-day origin dispatch and delivery dates. It does *not* contain real-time GPS telemetry, weather sensor logs, road traffic feeds, customs holds, or intermediate milestone scans. Consequently, intermediate stage breakdowns (Origin Dispatch, Main Line-Haul, Transfer Hub, Delivery) and disruption propagation timelines are **engineering simulations** designed to model supply chain mechanics, clearly distinguished from historical data.

2.2 Preprocessing & Sanitization Pipeline

An exploratory audit of the 2,000 raw shipment records detected data hygiene issues requiring remediation:

- **Chronological Inversions (16 records / 0.8%):** Shipments where `Delivery_Date` occurred prior to `Shipment_Date` (yielding negative durations down to -30 days) were removed as data entry errors.
- **Missing Delivery Dates (32 records / 1.6%):** Incomplete records marked delivered but lacking arrival timestamps were excluded from duration training.

- **Lost Cargo (45 records / 2.25%):** Shipments marked `Lost` were filtered out as aborted transits.
- **Weight Outliers:** Zero-weight anomalies were filtered, and extreme values capped at the 99.5th percentile (152.7 kg) to mitigate undue leverage on regression tree splits.

Final Validated Training Dataset: 1,800 clean records (90.0% retention).

2.3 Target Leakage Prevention

To ensure operational validity, features generated post-departure were strictly excluded:

- **Excluded Post-Hoc Fields:** `Delivery_Date`, `Status`, and `Cost` (settled after delivery).
- **Strictly Excluded Target Leakage:** The column `Transit_Days` identically matches (`Delivery_Date - Shipment_Date`) in the raw data. Feeding it into the model represents 100% target leakage, so it was strictly excluded.
- **Valid Pre-Shipment Features Retained:** `Origin_Warehouse`, `Destination`, `Carrier`, `Distance_miles`, `Weight_kg`, `Shipment_Month`, `Shipment_DayOfWeek`.

3. Model Performance & Operational Limitations

3.1 Evaluation Methodology

Models were benchmarked using a standard **80/20 train/test split** with a fixed seed (`random_state=42`):

- **Training Set:** 1,440 validated shipments
- **Holdout Test Set:** 360 shipments
- **Baseline Model:** Central-tendency median delivery duration of the training set (4.00 days / 96.0 hours).
- **ML Pipeline:** Scikit-Learn `Pipeline` coupling a `ColumnTransformer` (OneHotEncoder + StandardScaler) with a `RandomForestRegressor` (200 estimators, max depth 12).

3.2 Measured Holdout Benchmark Results

Metric	Median Baseline	Random Forest Pipeline	Net Difference
MAE (Days)	2.114 days	**2.016 days**	**-0.098 days**
MAE (Hours)	50.73 hours	**48.38 hours**	**-2.35 hours (4.6% error reduction)**
RMSE (Days)	3.359 days	**3.105 days**	**-0.254 days**
RMSE (Hours)	80.63 hours	**74.53 hours**	**-6.10 hours variance reduction**
R^2 Score	-0.0886	**+0.0699**	**+0.1585 gain**

3.3 Plain-English Metric Interpretation

- **MAE (Mean Absolute Error):** Represents the average absolute error magnitude. An MAE of 48.38 hours indicates that duration forecasts miss actual arrival by approximately 2 days on average.
- **RMSE (Root Mean Squared Error):** Penalizes large outlier errors more heavily due to squaring. At 74.53 hours (versus 80.63 hours for baseline), the Random Forest successfully dampens extreme forecast misses.

- **R^2 (Coefficient of Determination):** At **0.0699** (~7% variance explained), the model indicates that static pre-departure features weakly explain total duration variance, with ~93% driven by unobserved in-transit dynamics (roadway congestion, weather delays, terminal dwell times).

3.4 Signed Error Convention

Residuals are defined using the consistent signed error convention:

```
prediction_error_hours = (predicted_duration_days - actual_duration_days) * 24.0
```

- **Negative values:** Underprediction (model forecasted shorter delivery than actual transit).
- **Positive values:** Overprediction (model forecasted longer delivery than actual transit).

3.5 Transparent Discussion of Model Limitations

- **Modest Improvement:** While the Random Forest reliably beats the naive median baseline, it provides a modest gain (2.35 hours) rather than high precision. Outputs are operational estimates for planning buffers, not guaranteed commitments.
- **Calendar-Day Resolution:** Timestamps in the source data record calendar days without hours/minutes, introducing unavoidable discrete rounding noise.
- **Single Random Split:** A single random 80/20 partition does not establish temporal generalization across future operating seasons or newly introduced carrier corridors.
- **Proposed Future Work:** Temporal backtesting cross-validation, integration of live GPS/ELD telematics (Samsara/Geotab), EDI 214 milestone scan feeds, and real-time NOAA weather impedance data.

4. ETA Arithmetic & Contract SLA Risk Engine

All timestamps are strictly standardized to timezone-aware UTC:

```
ETA_UTC = Departure_Timestamp_UTC + Predicted_Duration_Hours
```

When a contractual delivery deadline (Target SLA) is specified:

```
Delay_Hours = (ETA_UTC - Expected_Arrival_UTC) / 3600.0
```

- **On Time / Early (≤ 0 hrs):** Delivery is projected on schedule or ahead of buffer.
- **Low Risk (0 to 12 hrs):** Minor expected delay within routine operating buffers.
- **Moderate Risk (12 to 24 hrs):** Delay exceeds standard buffer; priority tracking recommended.
- **High Risk (> 24 hrs):** Severe delay; imminent contractual SLA breach requiring active dispatch intervention.

5. Multi-Stage Disruption Propagation Simulator

Supply chains consist of connected legs: **Origin Dispatch (15%)**, **Main Line-Haul Transit (55%)**, **Transfer Hub Cross-Dock (20%)**, and **Final Delivery (10%)**.

The simulator implements a **guaranteed single cumulative delay ledger**:

```
Stage_Start[i] = Stage_End[i-1]
Stage_End[i] = Stage_Start[i] + NominalDuration[i] + InjectedDelay[i] -
RecoveredBuffer[i]
```

When a +6.0 hour delay is injected during Main Transit, the final delivery ETA shifts by *exactly +6.0 hours* (when no recovery buffer is active). The bottleneck is applied once and carried downstream across all remaining legs without recursive re-injection.

6. Operational Feature Importance

Aggregating one-hot tree split importances back to operational parent features indicates their empirical association with transit duration:

- **Transit Distance (~42%):** Primary structural foundation for line-haul movement.
- **Carrier Selection (~23%):** Operational speed variances between express networks (FedEx, UPS) vs. standard regional ground freight.
- **Cargo Weight (~16%):** Heavy consignments require pallet handling, reducing terminal cross-dock throughput speed.
- **Origin-Destination Corridor (~14%):** Metropolitan corridor density and hub route infrastructure.
- **Dispatch Day of Week / Month (~5%):** Weekend staging depot cycles and seasonal shipping volume shifts.
